

EUROPEAN ECONOMICS ANALYST

Euro Area—Higher Energy Prices Undo Recent Inflation Progress

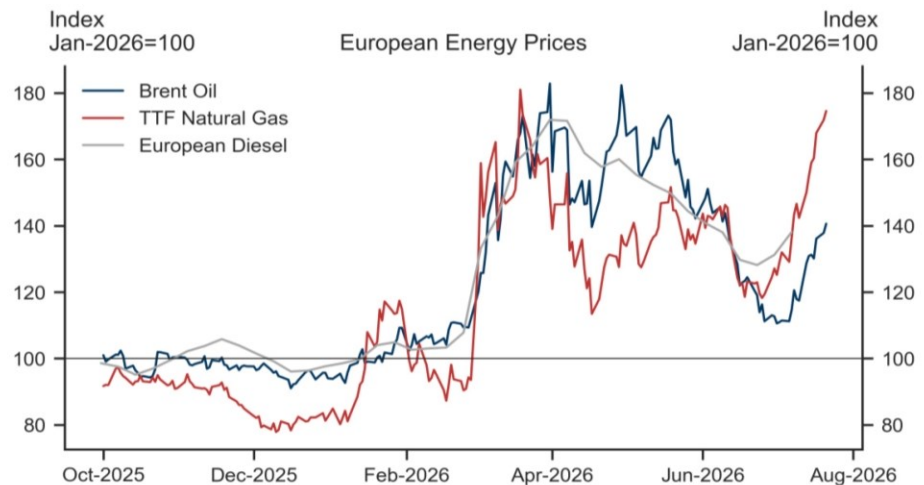
- The outlook for inflation in the Euro area is caught between opposing forces. On the one hand, the incoming inflation data has been weaker in Europe and globally, and price surveys have started to cool. On the other hand, the resumption of military conflict in the Middle East and Russian refinery outages have driven sharp increases in the prices of crude oil, oil products, and natural gas.
- We study the transmission of energy shocks along the inflation pipeline with a new statistical model (a large Bayesian Vector Auto-Regression). Our model jointly considers 19 variables such as producer price indices, shorter-term inflation expectations, consumer price inflation, and wage growth. This approach is similar to that used by central banks, including the ECB.
- Our model confirms that energy shocks usually have instantaneous and direct effects on consumer and producer energy prices. These effects then gradually appear in cost indicators, measures of shorter-term inflation expectations, and indirectly affected consumer prices, and can eventually lead to second-round effects on wage growth and longer-term inflation expectations.
- Our results show that direct effects were initially very strong in March, that some indirect effects had built up by April but faded by June, and that there were no convincing signs of second-round effects throughout.
- Our estimates suggest that, with the softer price data and energy forwards at their late-June lows, inflation had appeared on track to undershoot the ECB staff projections by 0.4pp for headline and around 0.2pp for core. However, the subsequent increase in energy prices has since brought our model's forecast for the inflation peak back in line with the staff's baseline. Our analysis therefore suggests that the renewed increase in energy prices has broadly undone the recent progress seen in price surveys and the inflation data.
- As a result, we continue to expect the Governing Council to hike a second time at its September meeting. While our base case is that the ECB then remains on hold, the risks are skewed towards further tightening if persistently higher energy prices cause the inflation outlook to deteriorate further.

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Euro Area—Higher Energy Prices Undo Recent Inflation Progress

The outlook for inflation in the Euro area is caught between opposing forces. On the one hand, the incoming inflation data has been weaker in Europe and globally, and price surveys—such as the PMIs and the European Commission’s—have started to cool. On the other hand, the resumption of military conflict in the Middle East and Russian refinery outages have driven renewed increases in the prices of crude oil, oil products, and natural gas (Exhibit 1).

Exhibit 1: Europe Is Facing Renewed Energy Price Pressures



Source: Goldman Sachs Global Investment Research, Haver Analytics

Keeping precise track of the inflation pressures in the pipeline is a difficult task. Our preferred approach for studying large and complex datasets of this kind is to use a Bayesian Vector Auto-Regressive (BVAR) model. The auto-regressive structure allows for rich dynamics between variables, while the Bayesian estimation approach improves the stability of the results.

We jointly model the 19 variables listed in Exhibit 2. These either relate to the energy shock, are price indicators subject to direct, indirect, or second-round effects, or serve as macroeconomic controls. Most of the price variables are expressed in seasonally adjusted, month-over-month terms and are observed through May or June. We include six lags of each variable, and the sample begins in 2003.

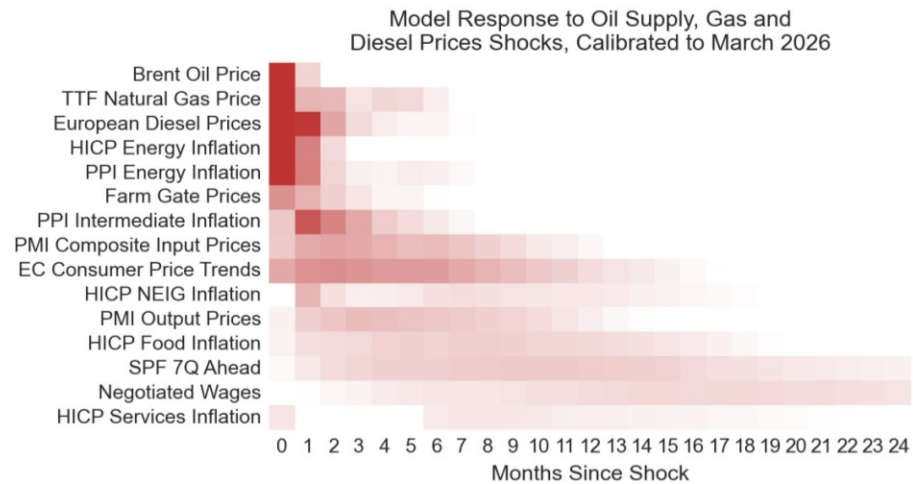
Exhibit 2: We Use a Bayesian Vector Auto-Regression to Jointly Model 19 Variables

Variable	Unit	Seasonally Adjusted	Last Observation	Category
World Oil Production	Percent, mom	No	July	Shock
Brent Oil Price	Percent, mom	No	July	Shock
TTF Natural Gas Price	Percent, mom	No	July	Shock
European Diesel Prices	Percent, mom	No	July	Shock
HICP Energy	Percent, mom	Yes	June	Direct
PPI Energy	Percent, mom	Yes	May	Direct
Farm Gate Prices	Percent, mom	No	June	Indirect
PPI Intermediate	Percent, mom	Yes	May	Indirect
PMI Composite Input Prices	Level	No	June	Indirect
12m Consumer Inflation Expectations	Level	No	June	Indirect
HICP NEIG	Percent, mom	Yes	June	Indirect
PMI Composite Output Prices	Level	No	June	Indirect
HICP Food	Percent, mom	No	June	Indirect
7q Mean Inflation Forecast	Percent, yoy	No	Q2	Second-round
Negotiated Wages	Percent, yoy	No	March	Second-round
HICP Services	Percent, mom	Yes	June	Second-round
PMI Composite Output	Index	No	June	Control
Unemployment Gap	Percentage Points	No	May	Control
Global Supply Chain Pressure Index	Index	No	June	Control

Source: Goldman Sachs Global Investment Research

We can visualise our model with a heatmap (similar to the one ECB Chief Economist Lane presented in a recent [speech](#)) showing the average strength and delay with which price indicators have historically responded to oil supply, gas price, and diesel price shocks (Exhibit 3). The variables are ordered from most upstream at the top to most downstream at the bottom; the time elapsed since the shock increases from left to right; and the colour gradient indicates the strength of the response.

Our estimates confirm that energy shocks have instantaneous and direct effects on consumer and producer energy prices. These effects then gradually appear in cost indicators, measures of shorter-term inflation expectations, and indirectly affected consumer prices, and can eventually lead to second-round effects on wage growth and longer-term inflation expectations.

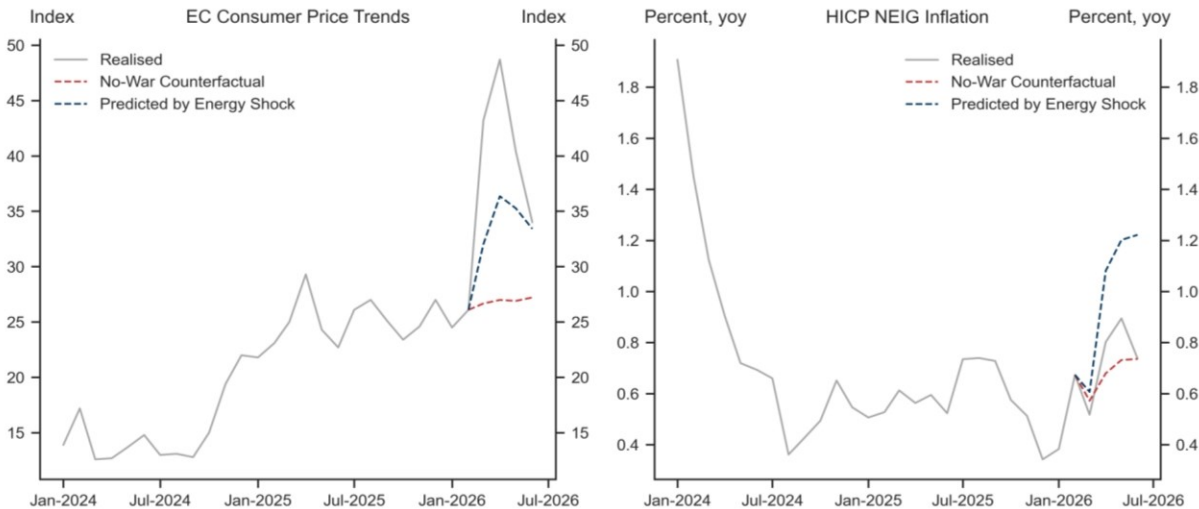
Exhibit 3: We Can Visualise Our Model With a Heatmap

The heatmap shows the standardised response of price indicators to the oil supply shock, gas price shock, and diesel price shock identified in March 2026. We use sign restriction on oil production and oil prices to identify oil supply shocks, and zero sign restrictions for gas price and diesel price shocks. Response at-or-above one standard deviations are shown with the darkest shade of red, while zero and negative responses are shown in white.

Source: Goldman Sachs Global Investment Research

We next use our model to ask how price variables would have evolved in the absence of war in the Middle East (obtained from the model's forecast before March) and if they had responded to the observed changes in oil production, as well as oil, gas, and diesel prices, in line with their historical behaviour.

We find that some price surveys (such as the European Commission's measure of shorter-term inflation expectations) initially reacted more strongly than usual but have since June returned in line with our model's prediction (Exhibit 4, left). In contrast, our model suggests that the response of non-energy industrial (or core) goods inflation has so far been more muted than in the past and no longer looks different from the no-war counterfactual (Exhibit 4, right).

Exhibit 4: The Response of Core Goods Inflation to Higher Energy Prices Has Been More Muted than Usual

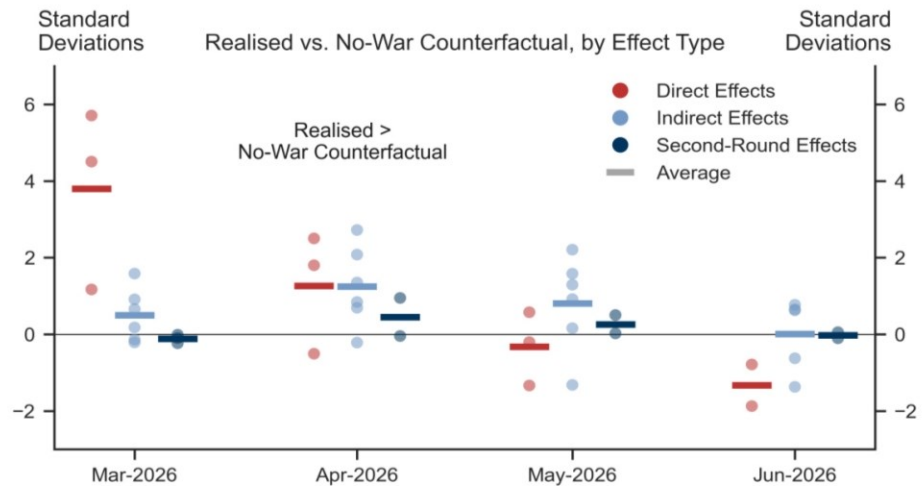
The no-war counterfactual is the model's unconditional forecast using realised data until February 2026. The path predicted by the energy shock is the model's forecast conditional on the realised data for oil production, oil prices, gas prices, and diesel prices between March and June 2026.

Source: Goldman Sachs Global Investment Research, Haver Analytics

The difference between the realised data and our model's no-war counterfactual can be interpreted as a measure of the strength of direct, indirect, or second-round effects, depending on the variable considered. We standardise this difference for every variable and plot the values as dots between March and June, with dashes indicating the average by type of effect (Exhibit 5).

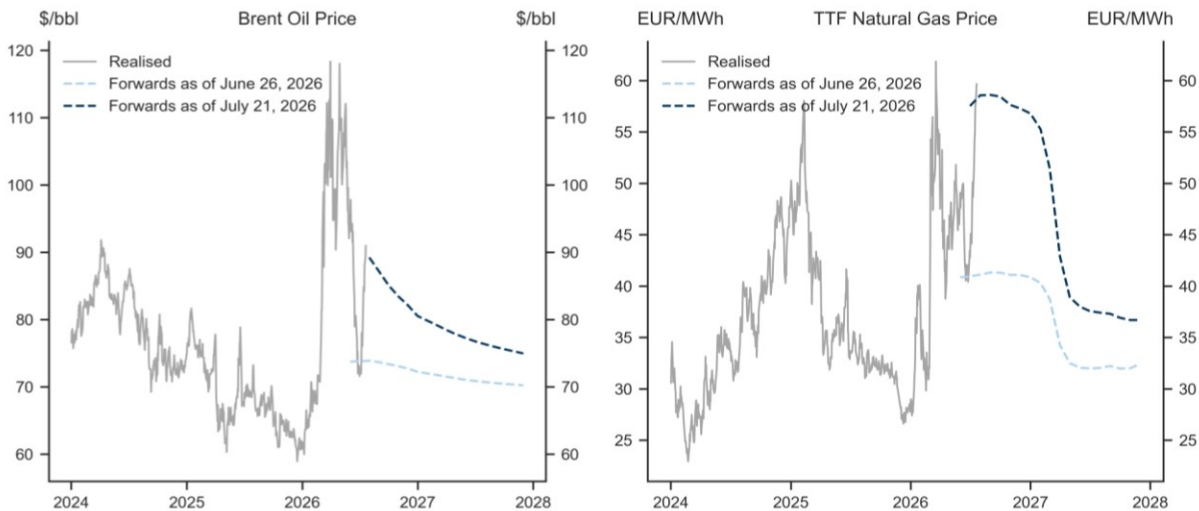
Our results show that direct effects were initially very strong in March, that some indirect effects had built up by April but faded by June, and that there were no convincing signs of second-round effects throughout.

Of course, the model itself would suggest that it is simply too early to detect signs of second-round effects at this stage. But in the case of wage growth, for instance, the decrease in one-year-ahead wage growth expectations reported by small and medium-sized firms in the ECB's SAFE survey is consistent with the absence of second-round effects so far.

Exhibit 5: Our Model Finds No Convincing Signs of Second-Round Effects So Far

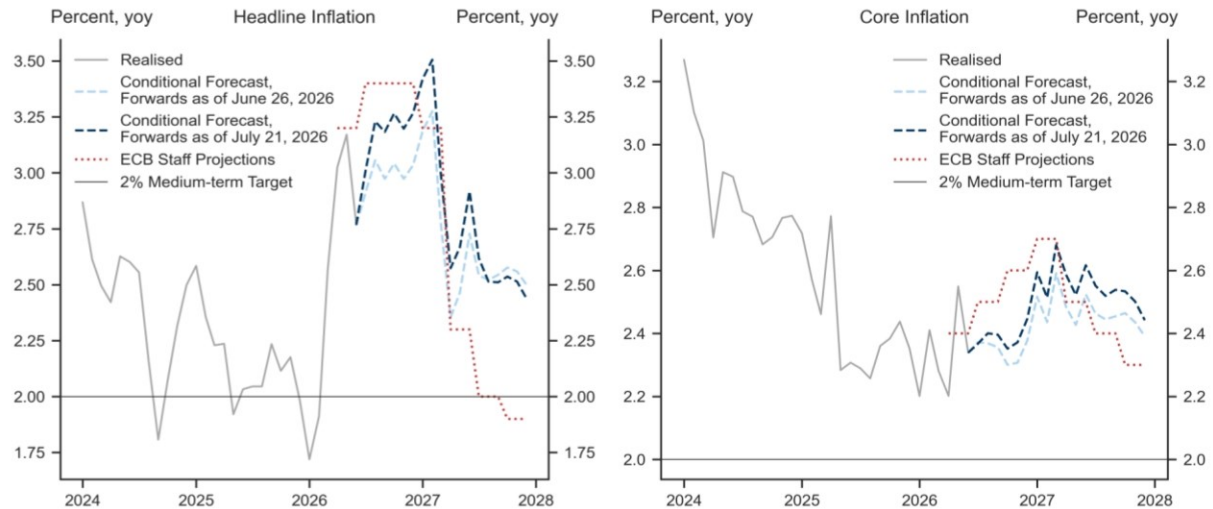
Source: Goldman Sachs Global Investment Research

Finally, we use our model to quantify the effect of higher energy prices on the inflation outlook. To do so, we first condition the model's forecast on energy forwards at their late-June lows (the light blue dashed lines in Exhibit 6) and then repeat the exercise with energy forwards at their current levels (dark blue).

Exhibit 6: We Condition Our Model's Forecast on Energy Forwards at Their Late-June Lows and at Their Current Levels

Source: Data compiled by Goldman Sachs Global Investment Research, Haver Analytics

The results for headline and core inflation are shown in Exhibit 7, with the June ECB staff projections included for comparison. Initially, with the softer price data and energy forwards at their lows, inflation appeared on track to undershoot the ECB staff projections by 0.4pp for headline and around 0.2pp for core. However, the subsequent increase in energy prices has brought our model's forecast for the inflation peak back in line with the staff's baseline.

Exhibit 7: The Increase in Energy Prices Has Brought Our Model's Forecast Back in Line With the ECB Staff Projections

The conditional forecasts use oil and gas price forwards from July and August 2026 until the end of 2027, while oil production and diesel prices evolve in line with the model.

Source: Goldman Sachs Global Investment Research, Haver Analytics, ECB

Our analysis therefore suggests that the renewed increase in energy prices has broadly undone the recent progress seen in price surveys and the inflation data. As a result, we continue to expect the Governing Council to hike a second time at the September meeting. While our base case is that the ECB then remains on hold, the risks are skewed towards further tightening if persistently higher energy prices cause the inflation outlook to deteriorate further.

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Disclosure Appendix

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